0. Brief introduction to quantum error correction (6.5 min)

0–1.5 min / 1 Quantum error correction Pauli errors, stabilizer codes, syndrome, $[\![n, k, d]\!]$ [Got97].

1.5–3.5 min / 2 CSS codes: two binary codes X errors: decode C1. Z errors: decode the dual of C2. Logical basis from S and K.

3.5–5 min / 3 Transversal gates and their limits Clifford plus T or CCZ is universal. Eastin–Knill: no code has a universal transversal set. Distillation supplies the missing gate.

5–6.5 min / 4 Codes with different transversal gates Steane: Clifford. 15 qubits: T. 8 qubits: CCZ. State the question.

a. The cube code (7 min)

6.5–7.5 min / 5 Qubits indexed by the points of $\mathbb{F}_2^3$ Binary vectors are functions on the points. Steps: u1, u2, u3, then the constant vector. They span RM(1,3).

7.5–8 min / 6 The cube code $[\![8, 3, 2]\!]$ K has rows u1, u2, u3, which is why qubits are points.

8–9 min / 7 Distances of the cube code Z000 Z100 commutes with the check and flips x1: d_Z = 2. The smallest interesting colour code.

9–9.5 min / 8 Codeword weight, physical phase, and logical phase T sites minus T-dagger sites. The gate condition mod 8.

9.5–11 min / 9 Choosing the signs on the cube Affine functions have 4 ones. p is the codeword of label 111. Define d_H.

11–13 min / 10 Reading the cube phase from overlaps Per-position parity identity. Face, edge, vertex: W = 0, 0, -1, so the phase is CCZ.

13–13.5 min / 11 The gate condition becomes conditions on generator rows Mod 8, mod 4, mod 2. Define the bracket.

b. History and the CH conditions (4 min)

13.5–14.5 min / 12 Historical context Cite the CSS-T work of Camps-Moreno, López, Matthews et al.

14.5–16 min / 13 Triorthogonal matrix and triorthogonal space Nezami-Haah puncture parent spaces. We fix S first and find K.

16–17.5 min / 14 Conditions for quasi-transversal CCZ Mod 2 reduction gives the nine parities. A physical layer needs the signed congruences.

c. Results and finite geometry (7.5 min)

17.5–19 min / 15 Result: a 48-qubit code with a transversal T/T† CCZ 39 to 46 excluded by Wills, Jain and Singh.

19–20 min / 16 Fix S using finite/projective geometry code Four steps. Only the logical rows are searched.

20–21 min / 17 Projectivity rules out weight-one and weight-two Z logicals Independent of the choice of logical rows.

21–22 min / 18 Complement of a projective subspace 63 points, remove the 15 with w = 00, three blocks of 16.

22–23.5 min / 19 Hyperplane sections and codeword weights u1 gives 24, w1 gives 32. Both divisible by 8.

23.5–25 min / 20 More geometric constructions The table of codes found. Counts are lower bounds except 48.

d. Column picture and search (8 min)

25–26.5 min / 21 Column indicators from the generator matrix Show G for the cube. chi_G = y because every x occurs with y = 1.

26.5–28 min / 22 The CH conditions are inner products with monomials Overlap parity is an inner product. Coset of RM(r-4, r).

28–29.5 min / 23 Logical rows in the column picture Logical label, T_K, T_S, the bracket. Step 2 shows K1.

29.5–31.5 min / 24 Finding the logical rows Algorithm 1: V_P, pairs, then Gaussian elimination for kappa 3. Witness gives d_Z = 3.

31.5–33 min / 25 Solving for the T/T† pattern Binary unknowns, linear over Z4. 26 T plus 22 T-dagger.

e. Bounds and implications (7 min)

33–35 min / 26 Stabilizer supports are Reed–Muller codewords Monomials of positive degree vanish at the origin.

35–36.5 min / 27 Our exclusion argument covers every length through 38 Analytic reductions plus a complete finite enumeration.

36.5–38.5 min / 28 Existence of a valid code at different block lengths Jacinto support is the affine version, not Q48 punctured.

38.5–40 min / 29 Takeaways and open questions Leave the QR codes up during questions.

40 minutes: finish and invite questions.